

Mingze Yuan

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EDUCATION

- **Harvard University** Boston, MA, United States
Ph.D. in Computer Science (Advisor: Prof. Na Li and Prof. Quanzheng Li) Sep 2024 - present
- **Peking University, Center for Data Science** Beijing, China
M.S. in Data Science (Advisor: Prof. Bin Dong) Sep 2021 - Jul 2024
- **Peking University, School of Mathematical Sciences** Beijing, China
B.S. in Statistics; Double degree in Economics Sep 2017 - Jul 2021

INDUSTRY EXPERIENCE

- **Medical AI R&D, Alibaba DAMO Academy** Beijing, China
Research Intern on AI for medical imaging Jun 2022 - Jun 2024

RESEARCH INTERESTS

My research interests include generative modeling and its applications in scientific domains, with a focus on AI for drug design. Additional expertise in image segmentation and medical image analysis.

SELECTED PUBLICATIONS

- **Mingze Yuan**, Pengfei Jin, Na Li, Quanzheng Li. *PIRF: Physics-Informed Reward Fine-Tuning for Diffusion Models*. Thirty-ninth Conference on Neural Information Processing Systems (**NeurIPS**) workshop on AI for science, 2025.
- Yiqi Tian*, Pengfei Jin*, **Mingze Yuan**, Na Li, Bo Zeng, Quanzheng Li. *RODS: Robust Optimization Inspired Diffusion Sampling for Detecting and Reducing Hallucination in Generative Models*. Thirty-ninth Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
- Xudong Yan*, **Mingze Yuan***, Yao Lu, Ying Zhang, Zifan Chen, Peng Bao, Zehua Li, Bin Dong, Li Yang, Li Zhang, Fangxu Zhou. *Versatile Stain Transfer in Histopathology Using a Unified Diffusion Framework*. IEEE International Symposium on Biomedical Imaging (**ISBI**), 2025.
- **Mingze Yuan**, Peng Bao, Jiajia Yuan, Yunhao Shen, Zifan Chen, Yi Xie, Jie Zhao, Quanzheng Li, Yang Chen, Li Zhang, Lin Shen, Bin Dong. *Large language models illuminate a progressive pathway to artificial intelligent healthcare assistant*. **Medicine Plus**, 2024.
- Jiajia Yuan*, Peng Bao*, Zifan Chen*, **Mingze Yuan***, Jie Zhao, Jiahua Pan, Yi Xie, Yanshuo Cao, Yakun Wang, Zhenghang Wang, Zhihao Lu, Xiaotian Zhang, Jian Li, Lei Ma, Yang Chen, Li Zhang, Lin Shen, Bin Dong. *Advanced prompting as a catalyst: Empowering large language models in the management of gastrointestinal cancers*. **The Innovation Medicine**, 2023. (*: these authors contributed equally)
- Yutong Xie, **Mingze Yuan**, Bin Dong, Quanzheng Li. *Unsupervised Image Denoising with Score Function*. Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS**), 2023.
- **Mingze Yuan**, Yingda Xia, Xin Chen, Jiawen Yao, Junli Wang, Mingyan Qiu, Hexin Dong, Jingren Zhou, Bin Dong, Le Lu, Li Zhang, Zaiyi Liu, Ling Zhang. *Cluster-Induced Mask Transformers for Effective Opportunistic Gastric Cancer Screening on Non-contrast CT Scans*. International Conference on Medical Image Computing and Computer-Assisted Intervention (**MICCAI**), 2023.
- **Mingze Yuan**, Yingda Xia, Hexin Dong, Zifan Chen, Jiawen Yao, Mingyan Qiu, Ke Yan, Xiaoli Yin, Yu Shi, Xin Chen, Zaiyi Liu, Bin Dong, Jingren Zhou, Le Lu, Ling Zhang, Li Zhang. *Devil is in the Queries: Advancing Mask Transformers for Real-world Medical Image Segmentation and Out-of-Distribution Localization*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023 **Highlight** (Top 2.5%).
- Hexin Dong*, Zifan Chen*, **Mingze Yuan**, Yutong Xie, Jie Zhao, Fei Yu, Bin Dong, Li Zhang. *Region-Aware Metric Learning for Open World Semantic Segmentation via Meta-Channel Aggregation*. The International Joint Conference on Artificial Intelligence (**IJCAI**), 2022.

MANUSCRIPTS UNDER REVIEW

- Boxiang Yun, **Mingze Yuan**, Yueyi Zhang, Tony C. W. Mok, Lina Lin, Guodong Jing, Yunhao Bai, Ning Zhang, Kai Zhang, Chao Ma, Xiaochen Feng, Jiacheng Yang, Qinghui Yu, Shuai Zhao, Zufeng Chen, Yu Luo, Tingting Zhang, Wei Tang, Yu Shi, Jianping Lu, Chenwei Shao, Yan Wang, Ling Zhang, Yingda Xia, Kai Cao. *High-fidelity virtual contrast-enhanced CT via generative artificial intelligence: Development and validation in pancreatic oncology*. Submitted to **Nature Biomedical Engineering** (under review).
- Yichun He, Hao Sheng, **Mingze Yuan**, Yiming Zhou, Hailing Shi, Wendy Xueyi Wang, Zefang Tang, Zuwan Lin, Wenbo Wang, Na Li, Jia Liu, Xiao Wang. *An agentic AI foundation model for a universal spatial molecular atlas of the mouse brain*. Submitted to **Nature** (revised and resubmitted).

ACADEMIC SERVICE

- Journal reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2024
- Conference reviewer: ICML 2023, RSS 2025, NeurIPS 2025

SELECTED HONORS AND AWARDS

- **Gold Medal**, 32nd China Mathematics Olympiad (CMO), 2016
- **China National Scholarship**, Peking University, 2023
- **Challenge Winner Award (Champion)**, CrossMoDA, MICCAI 2022

SKILLS SUMMARY

- **Programming Languages**: Python, C/C++, Matlab, R
- **Languages**: English (TOEFL: 104, R30, L27, S21, W26), Mandarin